MILITARY SPECIFICATION

RADIO SET AN/SRC-40

DIGITAL DATA TRANSMITTER TERMINAL

This specification has been approved by the Naval Air Systems Command, Department of the Navy

1.0 SCOPE

1.1 Scope - The Digital Data Transmitter Terminal, Radio Set AN/SRC-40, hereinafter referred to as the AN/SRC-40, shall provide simultaneous UHF/FSK transmission of the CAINS Alignment and Waypoint data to aircraft equipped with Link-4A airborne equipment equivalent to the AN/ASW-25A/B or AN/ASW-27B. The AN/SRC-40 shall also be capable of transmitting the ACLS data link signal as a single-channel transmitter. Messages transmitted by the AN/SRC-40 are of the UHF/FSK type as defined by message standards of WR-28 and this specification.

1.2 Associated Equipment - The AN/SRC-40 shall operate with the AN/SSW-1 or similar equipment providing the proper input time-division data link messages. The transmitter and frequency synthesizer modules utilized in this equipment are interchangeable with the assemblies from the SM-511/ASW. The receiver module utilized in the equipment is interchangeable with the AN/ASW-25 receiver module.

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2.0 APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

Specifications:

Military:

MIL-C-915	Cable, Cord, and Wire, Electrical
MIL-E-16400	Electronic Equipment, Naval Ship and Shore; General Specification
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclatures and Nameplates for Aeronautical Electronic and Associated Equipment

Naval Air Systems Command:

AR-57	Shipboard Data Link Requirements for Aircraft Internal Navigation System Alignment and Waypoint Insertion
WR-28	Message Standards for AN/USC-2, or Equiv., Time Division Data Transmission Systems
EI-600	Installation Instructions for Radio Set AN/SRC-40, Digital Data Transmitter Terminal

Standards:

Military:

MIL-STD-130	I.D. Marking of U.S. Military Property
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-794	Part and Equipment, Procedures for Packaging and Packing

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Standards: (Continued)

Military:

MIL-STD-781	Reliability Tests, Expotential Distribution
MS17322	Meter, Time Totalizing, 115 Volt, 400-Cycle, Rev B

2.1.1 Availability of Documents - When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Publications & Forms Center, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania, 19120.

3.0 REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(1) All Microelectronic Integrated-Circuit (MIC) elements are considered to be nonstandard parts. As such, their usage shall be reported to the procuring agency for approval.

(2) Other parts and materials requirements shall conform to MIL-E-16400.

(3) Nonrepairable subassemblies, as outlined in MIL-E-16400, shall be used when practicable.

(4) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable subassemblies without the approval of the procuring activity.

3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials shall be obtained as specified in MIL-E-16400.

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3.2.2 Modules, Maintenance - The electronic portions of the AN/SRC-40 shall be divided into removable modules. Each transmitter module shall be replaceable without the necessity to disconnect power from the AN/SRC-40.

3.3 Design and Construction - The AN/SRC-40 shall conform with the applicable requirements of MIL-E-16400 for design, construction, and workmanship for Class 4 equipment except as otherwise specified herein. Maximum use of existing design shall be utilized.

3.3.1 Total Weight - The total weight of the AN/SRC-40 shall be a minimum consistent with good design and shall not exceed 75 pounds, excluding external cable assemblies, antenna assembly, and Handbook.

3.3.2 Reliability

3.3.2.1 Operational Stability - The AN/SRC-40 shall operate with satisfactory performance, continuously or intermittently, for a period of at least 1000 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The AN/SRC-40 shall have a minimum total operating life of 50,000 hours with reasonable servicing and replacement of parts. A list of parts requiring scheduled replacement due to wear during the life of the AN/SRC-40, and the wearout life of such parts, shall be determined by the contractor and submitted to the procuring agency.

3.3.2.3 Reliability in Mean-Time-Between-Failure (MTBF) - The AN/SRC-40 shall have a minimum of 1,000 hours of mean operating time between failures when tested and accepted as outlined under the requirements of Paragraph 4.3.1 herein.

3.3.2.4 Time Totalizing Meter - The AN/SRC-40 shall contain a time totalizing meter in accordance with MS17322. It shall be mounted on the front panel.

3.3.3 Connectors - The AN/SRC-40 shall provide for the use of external connectors in accordance with MIL-E-16400.

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3.3.4 Control Panel - The AN/SRC-40 shall utilize front panel controls as specified in MIL-E-16400.

3.3.5 Interchangeability - The AN/SRC-40 shall meet the interchangeability requirements as defined in MIL-E-16400.

3.3.6 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of MIL-STD-461 except as specified in Paragraph 4.2.2 (10) herein.

3.3.7 Maintenance Provisions - Provisions for maintenance shall be as specified in MIL-E-16400.

3.3.8 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with MIL-N-18307.

3.3.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests:

(1)	Temperature:	Room ambient ($25^{\circ} \pm 10^{\circ}$ C)
(2)	Altitude:	Normal ground
(3)	Vibration:	None
(4)	Humidity:	Room ambient
(5)	Input Power Voltage:	115 ± 1.0V., 400 ± 5 Hz

3.3.10 Service Conditions - The AN/SRC-40 shall operate satisfactorily under any of the environmental service conditions specified in MIL-E-16400 for Class 4 equipment, except as otherwise specified herein.

3.3.10.1 Vibration - The AN/SRC-40 shall meet the vibration requirements of MIL-E-16400, except the maximum frequency of vibration shall be 33 Hz.

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3.3.10.2 Shock - The AN/SRC-40 shall not suffer damage or subsequently fail to provide the performance specified herein when subjected to 18 impact shocks of 20 G, consisting of 3 shocks in opposite directions along each of 3 perpendicular axes, with each shock impulse having a time duration of 11 ± 1 milliseconds. The "G" value shall be within ± 10% when measured using a 0.2 to 250 Hz filter, and the maximum "G" shall occur at approximately 5-1/2 milliseconds. The unit shall be operational after test.

3.3.10.3 Temperature - The AN/SRC-40 shall be capable of normal operation, as specified herein, when subjected to the temperature cycle requirements of MIL-E-16400.

3.3.10.4 Humidity - The AN/SRC-40 shall be capable of meeting the humidity test of MIL-E-16400.

3.3.11 Warm-up Time - The time required for the AN/SRC-40 to warm up prior to operations shall be kept to a minimum and shall not exceed 10 minutes under standard conditions and 20 minutes at extreme service conditions. The time delay is applicable only to obtaining specified frequency stability, as defined in Paragraphs 3.5.1.6.1(4) herein.

3.3.12 Primary Input Power Requirements - The AN/SRC-40 shall be capable of satisfactory operation when supplied input power having characteristics as defined in MIL-E-16400. The power required shall be of the following type:

AC Power, Single Phase, 115 V, 400 Hz, 200 VA

A means shall be provided to deenergize the entire AN/SRC-40 with one switching function. A power ON/OFF indicator shall also be provided. The capability to remotely energize and deenergize the AN/SRC-40 shall also be provided.

3.3.12.1 Primary Power Circuits and Overload Protection - The primary power circuits and overload protection shall meet the requirements of MIL-E-16400. Fuses shall not be utilized.

3.3.13 Cooling - The AN/SRC-40 shall operate within the ranges established in MIL-E-16400 with no external cooling provisions.

3.4 Performance - The performance of the equipment shall be as set forth below. Unless otherwise specified, these specifications shall apply under both standard and extreme service conditions. When reduced performance requirements for service conditions are acceptable, such variations will be specified.

3.5 Detail Requirements

3.5.1 Digital Data Transmitter Terminal - The AN/SRC-40 shall meet the following requirements.

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3.5.1.1 Function - The AN/SRC-40 shall have the capability of receiving two binary data streams containing CAINS Primary (Alignment) and Secondary (Waypoint) data. Two separate UHF carriers shall be modulated by the data streams and two separate UHF/FSK signals shall be provided as output signals.

Reception of these AN/SRC-40 output signals by aircraft equipped with Link-4A airborne equipment equivalent to the AN/ASW-25A/B or AN/ASW-27B will allow automatic Alignment and Waypoint data insertion for the on-board inertial navigation systems.

3.5.1.1.1 Transmitter Duty Cycle -

(1) In the CAINS Transmission Mode, the primary and secondary transmitters shall transmit a continuous wave (CW) carrier. Data link messages shall be transmitted at a rate of one message every 16 milliseconds in accordance with AR-57.

(2) In the ACLS Mode, the transmitter carrier shall alternately be turned on for 14 milliseconds and off for 18 milliseconds.

The AN/SRC-40 shall also be capable of operating as an All Weather Carrier Landing data link transmitter. In this mode the AN/SRC-40 shall transmit only a single UHF/FSK signal modulated by the ACLS data from AN/SSW-1( ) or equivalent.

3.5.1.1.2 Redundant Design - To ensure that the system shall be reliable to the fullest extent possible within economical limits, the AN/SRC-40 shall include a backup transmitter that will automatically be switched in to replace a failed transmitter. Redundancy shall be utilized as follows:

(1) Dual Channel Transmitter Mode - The backup transmitter shall be switched in to replace the primary or secondary transmitter in the event that UHF power output of either the primary or the secondary transmitters drops more than 4 ± 2 dB below the 5 watt level. This switching shall be accomplished in less than 16 milliseconds. In the event that failure occurs in both the primary and secondary transmitters, the primary channel shall have priority for the backup transmitter. Failure of the primary transmitter shall also be detected by checking for the presence of data at the output of the receiver. In the event that the power output of both the primary and backup transmitters is 4 ± 2 dB below the 5 watt level, then the secondary transmitter and frequency synthesizer shall be used to transmit the primary data. Automatic switchover of the input data and proper UHF frequency selection shall be made. This switching shall be performed in less than 32 milliseconds.

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3.5.1.1.2 (continued)

(2) Single Channel T/R Mode - In the single channel T/R mode. A fully redundant transmit channel shall be provided by switching in the secondary frequency synthesizer and the backup transmitter in the event the primary transmitter or primary synthesizer fails.

3.5.1.2 Form Factor - The equipment shall be housed in a Shelf-mountable case with captive hardware in the box bottom for attachment to a rack mounted slide-out shelf. The front panel shall be a standard 19-inch width and shall not exceed 10.5-inches in height.

3.5.1.2.1 Surface Finish - The surface finish of the equipment will be in accordance with MIL-E-16400.

3.5.1.3 Weight - The weight of the AN/SRC-40 shall be minimum consistent with good design and shall not exceed 75 pounds, excluding external cable assemblies, antenna assembly, and Handbook.

3.5.1.4 Contents - The equipment shall contain the following assemblies, sub-assemblies, functional circuits, etc.

(1) Electrical equipment housing assembly.

(2) Two frequency synthesizer assemblies.

(3) Three UHF transmitter assemblies.

(4) UHF receiver assembly.

(5) Power supply assembly.

(6) Front control panel assembly.

(7) Connector panel assembly.

(8) Printed circuit card assemblies.

(9) PF switching circuitry.

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3.5.1.5 Controls - The front panel of the AN/SRC-40 shall contain controls for operation of the AN/SRC-40. The following functions shall be provided:

(1) Power - On/Off/Remote
(2) Primary XMTR FREQ.
(3) Primary XMTR ATTEN.
(4) Secondary XMTR ATTEN.
(5) Mode Control
(6) Test Controls
(7) XMTR Reset Switch
(8) Manual Override Switch
(9) Circuit Breaker
(10) Elapsed Time Indicator
(11) Status Indicators
(12) Secondary Transmitter Modulation Test Switch (rear panel).

3.5.1.5.1 Power - On/Off/Remote - This function shall be provided by a three position all lever switch with pull-to-operate lock. The ON position is Power ON; the OFF position is Power Off; and the REMOTE position shall allow a remote switch to control the Power ON/OFF function.

3.5.1.5.2 Frequency Selection - Primary XMTR FREQ. - The frequency selection shall be accomplished by a series of thumbwheel switches properly indexed to provide 0.1 MHz tuning throughout the frequency range of 300.0 to 324.9 MHz. The most significant numerical digit may be engraved on the front panel to minimize the number of thumbwheel switches required to provide visual readout of the selected frequency.

3.5.1.5.3 Primary XMTR. ATTEN. - An RF attenuator shall be utilized to vary the Align transmitter output. The attenuator shall be adjustable over the range of -10 to -80 ± 2 dB in 10 dB steps. A manually operated coaxial switch shall be provided to switch the attenuator in or out of the RF line.

3.5.1.5.4 Secondary XMTR. ATTEN. - An RF attenuator shall be utilized to vary the Waypoint output power from -10 to -80 ± 2 dB in 10 dB steps. A manually operated coaxial switch shall be provided to switch the attenuator in or out of the RF line.

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3.5.1.5.5 Mode Control - Pri./Sec. Chan. XMIT only/Pri. Chan. XMIT/REC - A mode control shall be provided to switch the transmitter terminal from use as a CAINS Primary and Secondary data transmitter to a single channel ACLS data transmitter. This control shall provide for external keying of the transmitter and shall utilize the ACLS data input in place of the Align data input for modulation of the Align transmitter.

3.5.1.5.6 Test Controls - To provide the capability to simulate transmitter failure, three toggle type test switches shall be provided. These switches shall be used to turn off the carrier of each of the three internal transmitters. A momentary type toggle switch shall be used to prevent accidental operation of these three switches. A test switch for inhibiting the primary XMTR modulation fault detection logic shall also be provided.

3.5.1.5.7 XMTR Reset - There shall be a press-to-reset control for recycling the failure detect and priority logic. This control will allow the primary and secondary transmitters to be placed back into operation after having been switched to the OFF position by a failure.

3.5.1.5.8 Manual Override Switch - A six (6) position rotary switch shall be used to force the system into the following modes of operation:

(1) B/U XMTR OFF
(2) Secondary and B/U XMTR's OFF
(3) Secondary XMTR OFF
(4) Normal
(5) Primary XMTR OFF
(6) Primary and B/U XMTR OFF

3.5.1.5.9 Circuit Breaker - To provide overall circuit protection, a circuit breaker shall be provided. This circuit breaker shall be resettable and conveniently located.

3.5.1.5.10 Elapsed Time Indicator - An elapsed time indicator shall be provided on the control panel to record operating time.

3.5.1.5.11 Status Indicators - Indicators shall be located on the front panel of the unit as follows:

(1) GO/NO-GO Indicators
- (a) Primary Transmitter
- (b) Primary Data Input
- (c) Secondary Transmitter
- (d) Secondary Data Input
- (e) Backup Transmitter

(2) Monitor Indicators
- (a) Power ON
- (b) Backup XMTR Standby
- (c) Manual Override ON

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3.5.1.5.11 (continued)

Both the GO and NO-GO lights for the Backup transmitter shall be deenergized until the Backup transmitter is energized for use due to a failure of the Waypoint or Align transmitters. In the ACLS Mode, the GO and NO/GO lights for the Secondary XMTR data input shall be deenergized.

3.5.1.5.12 Sec. XMTR MOD TEST Switch - A switch shall be provided to switch the receiver L.O. input from the primary frequency synthesizer to the secondary frequency synthesizer for test purposes. This switch need not be on the front panel.

3.5.1.5.13 Protective Cover - The left half of the front panel shall be protected against inadvertent operation by a hinged cover which shall be secured at the bottom and swing upward. All indicator lights will be exposed.

3.5.1.6 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Connector Type	Function
J1	MS3112E-12-3P	Primary Power
J2	MS3112E-10-41P	Remote Indicators
J3	MS3112E-12-10P	Data Input
J4	TNC M39012/28-0001	PRI. R.F. OUTPUT
J5	TNC M39012/28-0001	SEC. R.F. OUTPUT
J6	MS3112E-22-55S	Test Points
* J7	TNC M39012/28-0001	Primary Atten R.F. OUTPUT
* J8	TNC M39012/28-0001	XMIT/REC R.F. INPUT

3.5.1.6.1 Output Signals - The AN/SRC-40 shall provide the following minimum outputs for external use.

3.5.1.6.1.1 UHF/FSK Primary Signal - An internal UHF/FSK transmitter and associated front panel control shall be provided as follows:

(1) Frequency Channel - By means of front panel controls entitled PRI XMTR FREQ it shall be possible to select any frequency channel in the range of 300.00 MHz to 324.9 MHz in 100 KHz increments. Main channel frequencies shall correspond with the arithmetic series 300.0, 300.1, 300.2, 300.3,..324.9 MHz.

*For normal operation a jumper must be in place between these two connectors.

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3.5.1.6.1.1 (continued)

(2) RF Power Output - A UHF signal of 5 watts +2 -1 dB shall be provided by the transmitter modules. A series adjustable RF attenuator shall be contained to allow attenuation of the signal by 80 ±2 dB. When the attenuator is switched out the insertion loss between the transmitter module output and the AN/SRC-40 output connector shall not be greater than 1.5 dB.

(3) RF Load Impedance - The transmitter shall operate as specified when terminated in 50 ohm resistive load. No damage shall result to the transmitter when terminated in an open or short circuit or any impedance between these extremes.

(4) Frequency Stability - The UHF output signal frequency stability shall be ±2.5 kHz maximum when the equipment is operated under Standard Conditions as specified herein and ±3.5 kHz maximum when operated at the temperature extremes specified in MIL-E-16400 for Class 4 equipment.

(5) Modulation Characteristics - The signal of Paragraph 3.5.1.6.1.1 (2) shall be FSK modulated by the signal of Paragraph 3.5.1.6.2.1 or 3.5.1.6.2.3. A binary one applied to the transmitter input shall cause the carrier to increase by 20 ±1 kHz. A binary zero shall cause the frequency to decrease by 20 ±1 kHz.

(6) Residual FM - Without true modulation, the frequency deviation developed by internal noise, including power supply impulse, shall not exceed 1 kHz peak-to-peak deviation when measured over a frequency range of 20 Hz to 8 kHz.

(7) Spurious Radiation - The magnitude of all radiated signals other than the carrier frequency and harmonics shall be controlled within the limits specified in Figure 1. The second harmonic of the carrier signal shall be 50 dB or more below the carrier. The third and higher harmonics shall be 75 dB or more below the carrier.

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3.5.1.6.1.1 (continued)

The spurious output levels shall be controlled as follows:

Spurious Frequency	dB (minimum) Below Carrier
150 KHz to 225 MHz	75
225 MHz to 300 MHz	75 to 50 at a slope corresponding to 0.33 dB increase per MHz
300 MHz to 325 MHz	50
325 MHz to 400 MHz	50 to 75 at a slope corresponding to 0.33 dB decrease per MHz
400 MHz to 10 GHz	75

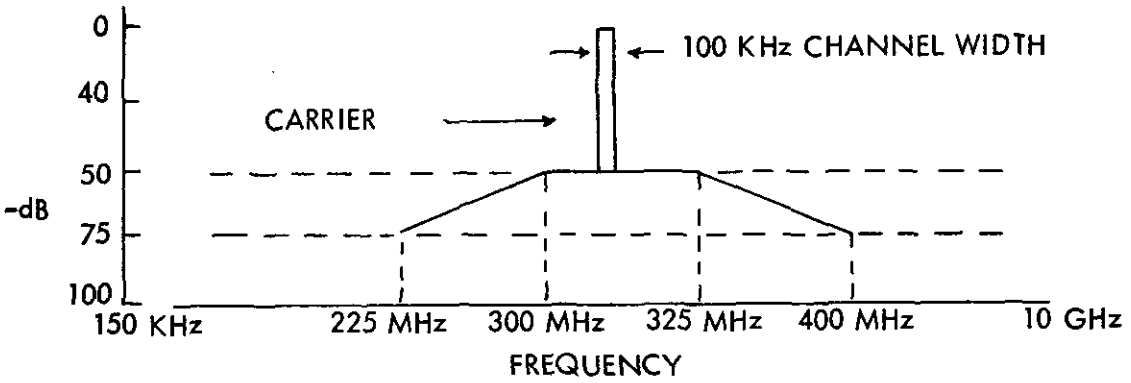


Figure 1. Limits of Spurious Radiation

(8) RF Output Control - In the ACLS mode the transmitter shall be controlled by a carrier ON/OFF control signal. The RF Power at the input to the receiver shall be less than -110 dBm when this signal is in the off condition.

3.5.1.6.1.2 UHF/FSK Seconday Signal - An internal UHF/FSK transmitter and associated front panel controls shall be provided as follows:

(1) Frequency Channels - The secondary transmitter shall use the same front panel channel frequency selection control as the primary transmitter. The secondary transmitter frequency will, however, be offset from the primary transmitter frequency by ±0.2 MHz. The resultant UHF channel frequency shall be 0.2 MHz above the primary channel when the tenths megahertz decade frequency of 3XX.0, 3XX.1, 3XX.4, 3XX.5, 3XX.8, or 3XX.9 have been selected. The resultant UHF channel frequency shall be 0.2 MHz below the primary channel when the tenths MHz decade frequency of 3XX.2, 3XX.3, 3XX.6, or 3XX.7 have been selected.

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3.5.1.6.1.2 (continued)

(2) Other Waypoint Transmitter Characteristics - The Waypoint transmitter shall also meet all the performance requirements defined in Paragraphs 3.5.1.6.1.1(2) through 3.5.1.6.1.1(8).

3.5.1.6.1.3 ACLS Reply Message - ACLS reply message signals occur during receive intervals as a result of replies from previously addressed aircraft. During transmit frames, the receiver picks up transmitted control messages which shall be used to compare with the input ACLS data to develop the Radio-Error signal. A Transmit/Receive switch shall be contained in the AN/SCR-40 to allow coupling of the transmitter to the align channel antenna during the carrier-on period and coupling of the receiver to the antenna during the carrier-off period. The AN/SRC-40 shall use the AN/ASW-25B receiver module, which shall meet the following requirements:

(1) Frequency Channels - The receiver shall be tuneable to any one of 250 channels in the range 300.0 MHz through 324.9 MHz in 0.1 MHz increments. The receiver tuning shall be by means of utilizing a portion of the Align transmitter frequency synthesizer power for the receiver first IF local oscillator.

(2) Intermediate Frequencies - The receiver shall be a superhetrodyne double conversion design. The first intermediate frequency shall be 70.0 MHz.

(3) RF Input Impedance - The impedance at the receiver RF input terminals shall be 52 ohms nominal.

(4) RF Input Signal Power - An RF input signal at the receiver RF input terminal from -100 dBm to +14 dBm shall produce proper data output as described in 3.5.1.6.1.3(11). Normal specified receiver sensitivity shall be obtained within one (1) millisecond after application of a +14 dBm signal at the input. An input signal power of +15 dBm shall not damage the receiver.

(5) Sensitivity - A -100 dBm (maximum) RF input signal at the selected channel frequency shall produce 20 dB output noise quieting measured on the output of a low pass filter connected to the receiver data output. The low pass filter shall have a 3 dB cutoff frequency of 5 ± 1 KHz and a minimum roll off of 18 dB per octave.

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3.5.1.6.1.3 (continued)

(6) Selectivity - The overall selectivity of the receiver shall fall within the limits below:

Attenuation	Bandwidth About Center Frequency
3 dB	± 35 KHz minimum
60 dB	+ 70 KHz maximum

Peak-to-valley ratio shall be 2 dB or less, plus 1 dB slope over a 70 KHz bandwidth.

(7) Frequency Stability and Tuning Accuracy - Overall drift of the receiver center frequency due to operation within the service conditions and as a result of component aging for a period of one (1) year shall not exceed ±1.2 KHz. This drift is exclusive of any drift caused by the external local oscillator signal. Internal tuning adjustments shall be provided to permit compensation for aging effects over a useful service life of 50,000 operating hours.

(8) Spurious and Image Response - When receiving a signal within the frequency range specified in Paragraph 3.5.1.6.1.3 (1), the receiver shall attenuate images and other spurious responses to the maximum practicable extent and in any case not less than -50 dB relative to the desired signal, except that only two (2) spurious responses per channel are permitted with attenuation between -45 and -50 dB.

(9) Local Oscillator Conduction - The maximum level of the local oscillator signal appearing at the receiver RF input terminals shall not exceed -67 dBm.

(10) Radar Environmental Compatibility - The receiver shall be capable of withstanding the following high power RF pulse input without damage:

Frequency Range:	200 to 225 MHz
Peak RF Power:	200 watts
Pulse Width:	10 microseconds
Pulse Repetition Rate:	400 pps

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(11) Reply Data Output Parameters - The reply data output shall be capable of driving a length of cable not to exceed 600 feet. Cable type will be TTRSA-10 cable per MIL-C-915. A mark is represented by a binary one and defined as 0.0 to -0.5 volts and a space is represented by a binary zero and defined as -3.0 to -4.5 volts measured at the end of the above 600 foot line. The signal source impedance shall be less than 50 ohms at a frequency of 100 KHz when measured into a capacity of 20,000 micromicrofarads which simulates the capacity of the above 600 foot line.

3.5.1.6.1.4 Remote GO/NO-GO Alarm Signals - Six sets of relay contacts (SPDT) shall be contained with input-output connections for remote GO/NO-GO indications and alarm for each of the following six conditions. Contact capacity of the GO/NO-GO contacts shall be 500 mA at 115 VAC. (See Figure 2).

(1) Primary Transmitter - Whenever the primary transmitter power output has deteriorated by 4 ±2dB or more, from 5 watts, or a modulation fault has been detected, the primary transmitter relay common line shall switch from the normally closed contact to the normally open contact (relay energized).

(2) Primary Data Input - Whenever the primary data input is not present, the primary data relay common line shall be switched from the normally closed contact to the normally open contact (relay energized).

(3) Secondary Transmitter - Whenever the secondary transmitter power output has deteriorated by 4 ±2dB or more from 5 watts, the secondary transmitter relay common line shall be switched from the normally open contact to the normally closed contact (relay deenergized).

(4) Secondary Data Input - Whenever the secondary data input is not present, the secondary data relay common line shall be switched from the normally closed contact to the normally open contact (relay energized).

(5) Backup Transmitter - The backup transmitter shall also contain a set of GO/NO-GO contacts which shall switch from the normally open contact to the normally closed contact whenever the power output deteriorates by 4 ±2dB or more when the backup transmitter has replaced the secondary transmitter, or whenever the power output deteriorates by 4 ±2dB or more or a modulation fault is detected when the backup transmitter has replaced the primary transmitter. The common lead of this set of contacts shall be in series with a second set of contacts which will be normally open except when either the primary or secondary transmitter are in a NO-GO condition. A Backup Standby light may be operated from the normally closed contact of the second set of contacts as shown in Figure 2.1 and 2.2.

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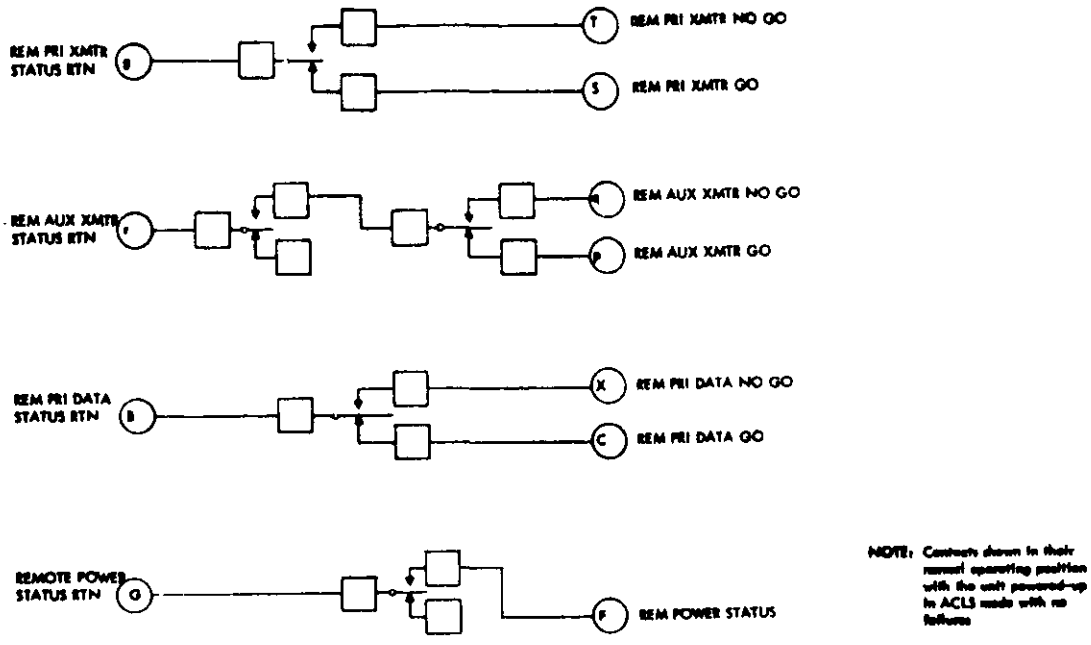


Figure 2-1. Connector J-2 External Alarm Interface Configuration, ACLS Mode

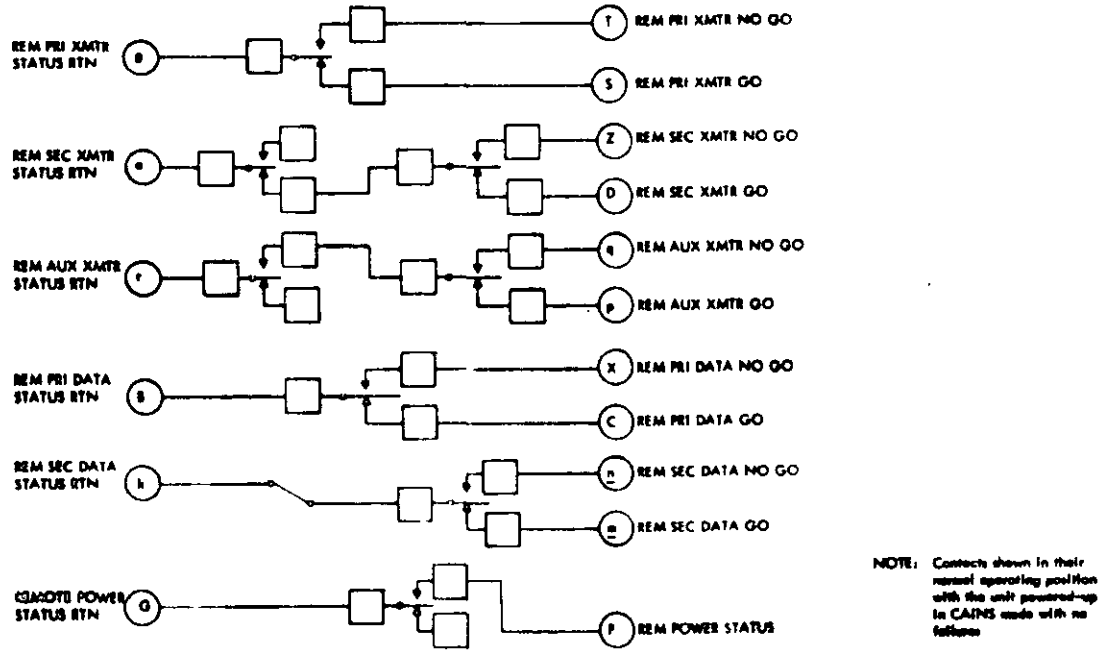


Figure 2-2. Connector J-2 External Alarm Interface Configuration, CAINS Mode

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(6) Power ON - A set of contacts to indicate POWER ON shall be contained in the AN/SRC-40. Contacts shall be closed when power to the unit is ON. Contacts shall be open when the power is OFF.

(7) Radio Error

(1) The AN/SRC-40 shall be capable of detecting a transmission error rate of at least 1 error per message. The radio equipment shall apply a positive 6 volts, 40 milliamperes maximum signal to an external line to indicate a radio error.

(2) The AN/SSW-1( ) Terminal shall provide an indicator lamp, designated Radio Error, which shall light upon receipt of a radio error signal as defined above.

(3) The detection of an error shall be defined as the transmission of any digit from time slot 13 through time slot 69 being the converse of the corresponding digit in the control message from the AN/SSW-1( ) Terminal. Upon receipt of an error in any message, the error signal shall be "locked-on" until manually released. The AN/SSW-1( ) shall provide a normally closed momentary operate switch in series with the Radio Error Indication lamp for the purpose of resetting the Radio Error Indication.

3.5.1.6.2 Input Signals - The AN/SRC-40 shall contain provisions for the following input signals.

3.5.1.6.2.1 Primary Data Input - The primary data input will be a serial time-division data link message stream of ±5V NRZ data. A Mark or One condition shall be represented by +5V +1V -0V, and a Space or Zero condition shall be represented by -5V -1V +0V. The input messages including synchronization shall have the message structure, timing, and frequency stability as defined by AR-57. The signal shall exhibit the above characteristics when terminated in a minimum load impedance of 2,000 ohms resistive. The input shall be protected against voltage transients of ±20 volts. Transitions from a Mark to a Space or a Space to a Mark shall not exceed 55 microseconds.

3.5.1.6.2.2 Secondary Data Input - The characteristics of this input signal will be identical to that of the primary data as specified in Paragraph 3.5.1.6.2.1 above.

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3.5.1.6.2.3 ACLS Control Message - The ACLS Control Message Input (CMSG) will be a serial time-division data link message stream. This signal is generated from a source impedance of less than 50 ohms at a frequency of 100 KHz when measured into a capacity of 20,000 micromicrofarads which simulates the capacity of a 600 foot line. The receiving amplifier shall be capable of accepting a binary one which is defined as 0.0 to -0.5 volts and a binary zero which is defined as -3.0 to -4.5 volts. The input current to the amplifier shall not exceed 3.0 milliamperes. The amplifier shall be capable of crosstalk and noise rejection of any signal having a peak-to-peak amplitude of not more than 1.2 volts when driven from a low impedance source over a 600 foot cable. The cable type is TTRSA-10 cable per MIL-C-915. All shields for each pair shall be grounded at the signal source end only.

3.5.1.6.2.4 Radio Sync - A binary zero shall occur on the Radio Sync line at the beginning of each data time slot in the control message starting with time slot 13. A binary one is defined as 0.0 to -0.5 volts. A binary zero is defined as -3.0 to -4.5 volts. The Radio sync pulses shall have a width of 4 microseconds or more as measured at the half amplitude points, at the AN/SRC-40 end of the 600 ft. cable.

The leading edge of any CMSG pulse, excluding the first 12 time slots, will occur within one microsecond of the leading edge of the corresponding Radio sync pulse. The additional characteristics of the Radio sync pulse are as defined in Paragraph 3.5.1.6.2.3.

3.5.1.6.2.5 CAR (Carrier) -

(1) The ACLS data transmitter shall turn on its carrier when the AN/SSW-1( ) applies a binary zero to the CAR line. A binary zero is defined as -3.0 to -4.5 volts. The radio transmitter shall turn off its carrier when a binary one occurs on the CAR line. A binary one is defined as 0.0 to -0.5 volt .

(2) The total interval from the time that the AN/SSW-1( ) applies a binary one on the carrier line until the transmitter has turned off its carrier and the receiver has recovered full sensitivity shall not exceed 1 millisecond.

(3) The radio transmitter shall be designed such that, if the carrier line from the AN/SSW-1( )Terminal should become open or become short circuited, the carrier shall turn off and remain off within 500 milliseconds after the last transition. The minimum time to switch off the carrier shall be 64 milliseconds.

(4) The additional characteristics of the CAR signal are as defined in Paragraph 3.5.1.6.2.3.

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4.0 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - The AN/SRC-40 shall be subjected to the following tests to determine conformance with all the applicable requirements.

(1) Preproduction (first article) Tests
(2) Acceptance Tests

4.2 Preproduction (first article) Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of 4.4. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction (first article) Test Data - The contractor shall submit all data collected in conducting these tests to the procuring activity for review and approval. On first production contracts, this data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and be obtained under the extreme high temperature operating condition.

4.2.2 Scope of Tests - Preproduction Tests shall include all tests deemed necessary to determine that the AN/SRC-40 meets all requirements of the contract and this specification. Preproduction Tests shall include environmental tests in accordance with the procedures of specification MIL-E-16400 and interference tests in accordance with specification MIL-STD-461 as detailed below:

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	Test	Requirements
(1)	Surface Examination	MIL-E-16400.
(2)	Operating Test	MIL-E-16400.
(3)	Weights and Dimensions	MIL-E-16400.
(4)	Supply Line Voltage and Frequency	MIL-E-16400, to the levels specified herein
(5)	Temperature	MIL-E-16400.
(6)	Humidity	MIL-E-16400.
(7)	Power	MIL-E-16400, except that power factor need not be determined. (The Power Test may be performed in conjunction with the Operating Test.)
(8)	Controls and Control Circuits	MIL-E-16400, successful completion of the Operating Test shall be sufficient to demonstrate suitability with no separate test being required.
(9)	Shock and Vibration	MIL-E-16400, (except 4 to 33 Hz only) for Vibration, and Paragraph 3.3.10.2 herein for Shock.
(10)	EMI Emission and Susceptibility	MIL-STD-461 except tests will be limited to the following:

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MIL-STD-461 Test Designation	Frequency Range	Limits
CE03 Power Line Conducted	.15 to 50 MHz	See Sheets 23 & 24
CE04 Signal & Control Line Conducted	.15 to 50 MHz	See Sheets 23 & 24
CS01 Power Line Conducted Susceptibility	.03 to 50 kHz	as stated in 461
CS02 Power Line Conducted Susceptibility	.05 to 400 MHz	as stated in 461
RE02 Electric Field Radiated	.15 MHz to 1 GHz	See Sheets 25 & 26
RS03 Electric Field Susceptibility	.15 MHz to 1 GHz	0.1 Volt into antenna

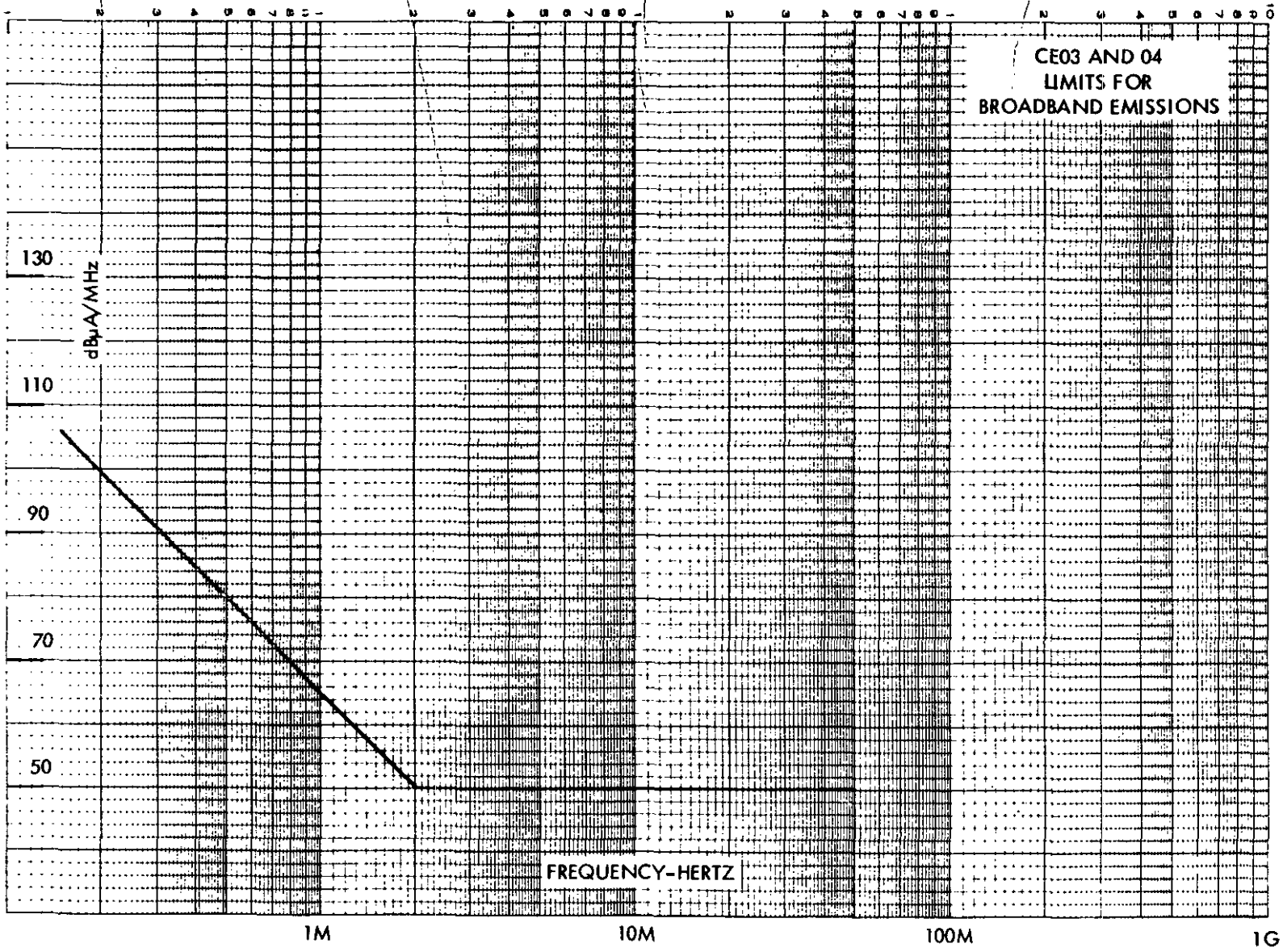
These tests shall be conducted after satisfactory completion of tests (1) through (9) above.

4.2.3 Preproduction (first article) Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted to acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

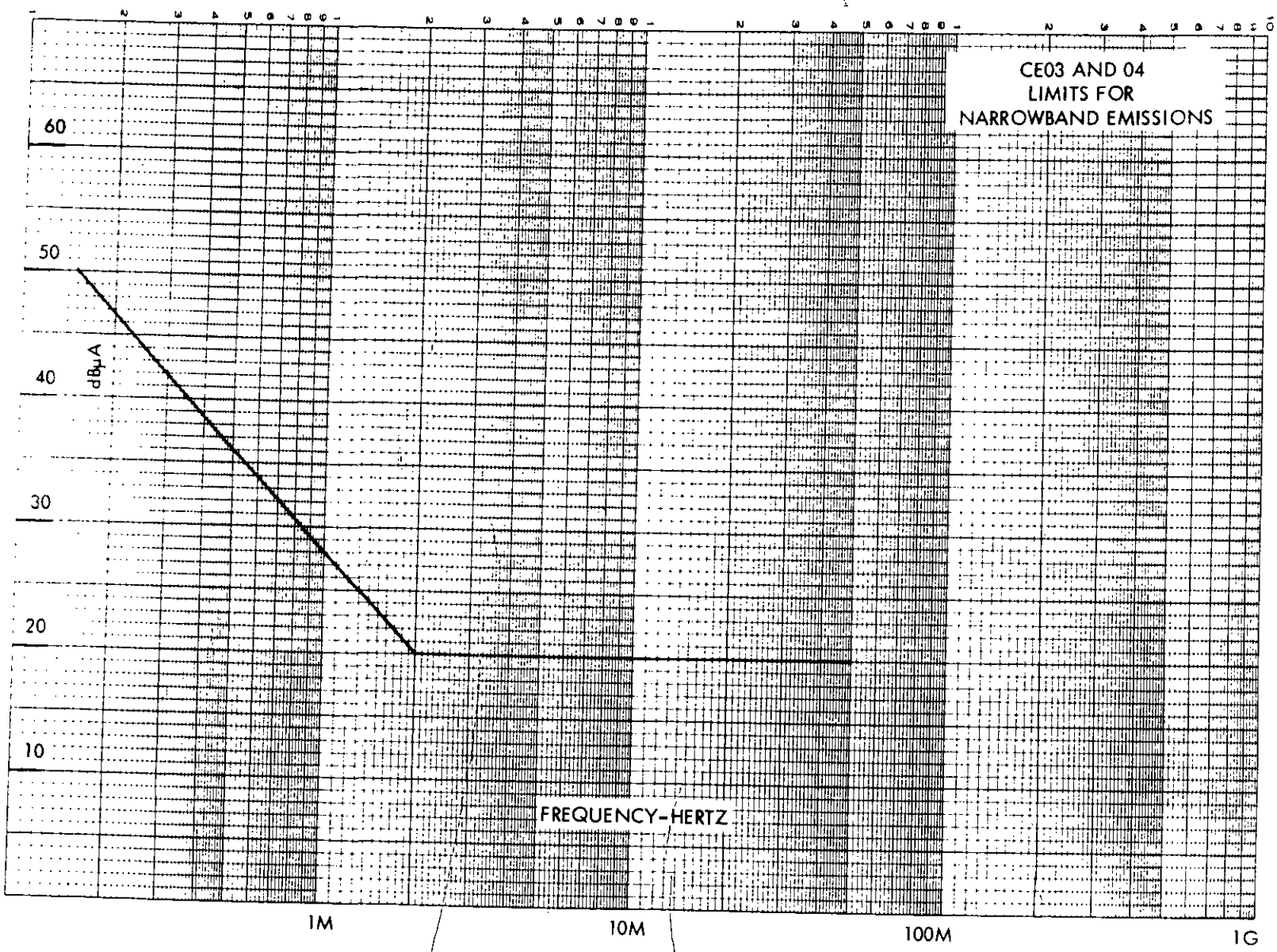
4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the Acceptance Tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the Procuring Activity shall engage the service of a commercial testing laboratory acceptable to the Procuring Activity. The contractor shall furnish test reports showing quantitative results for all Acceptance Tests required. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance Tests shall consist of the following:

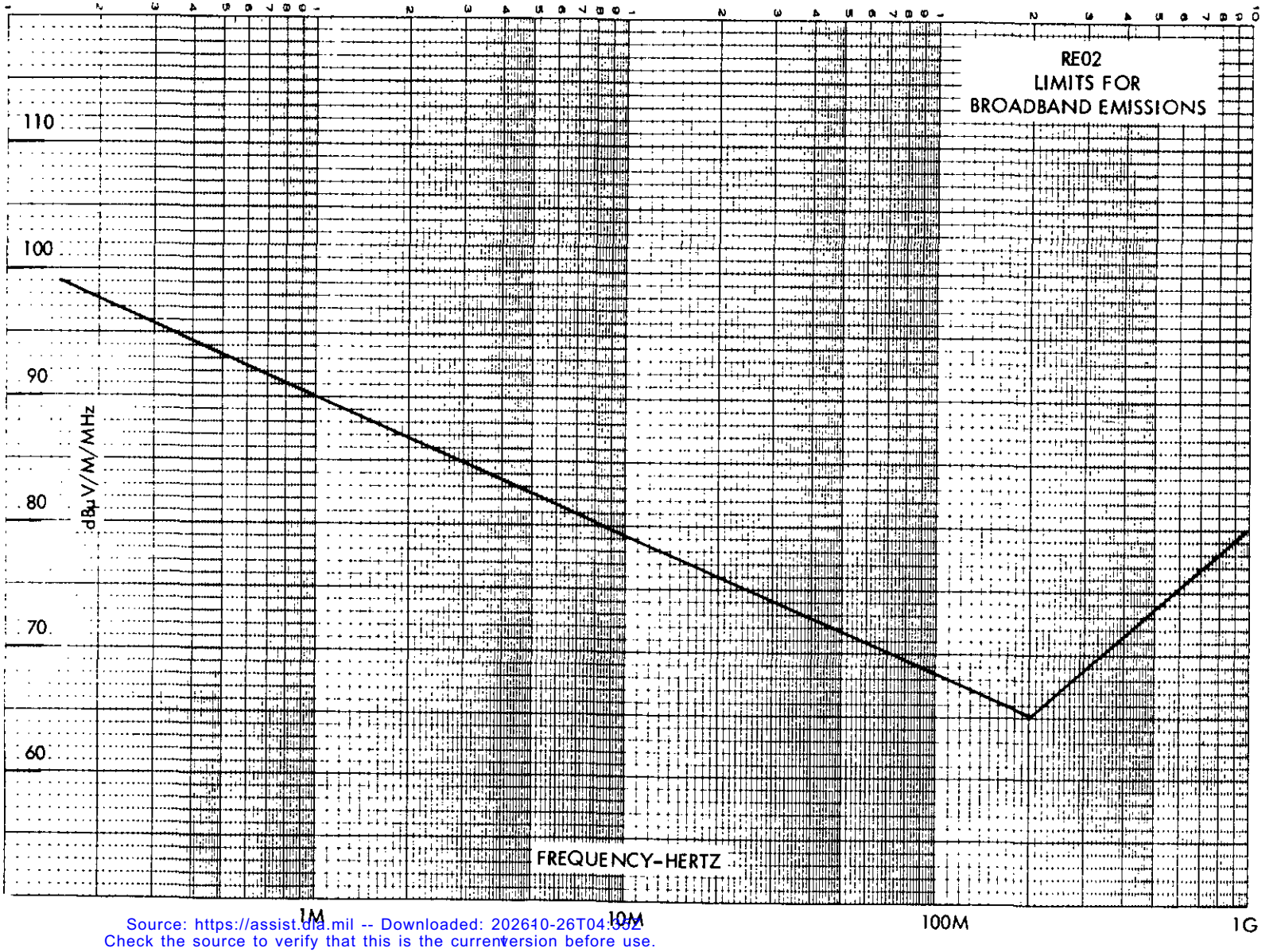
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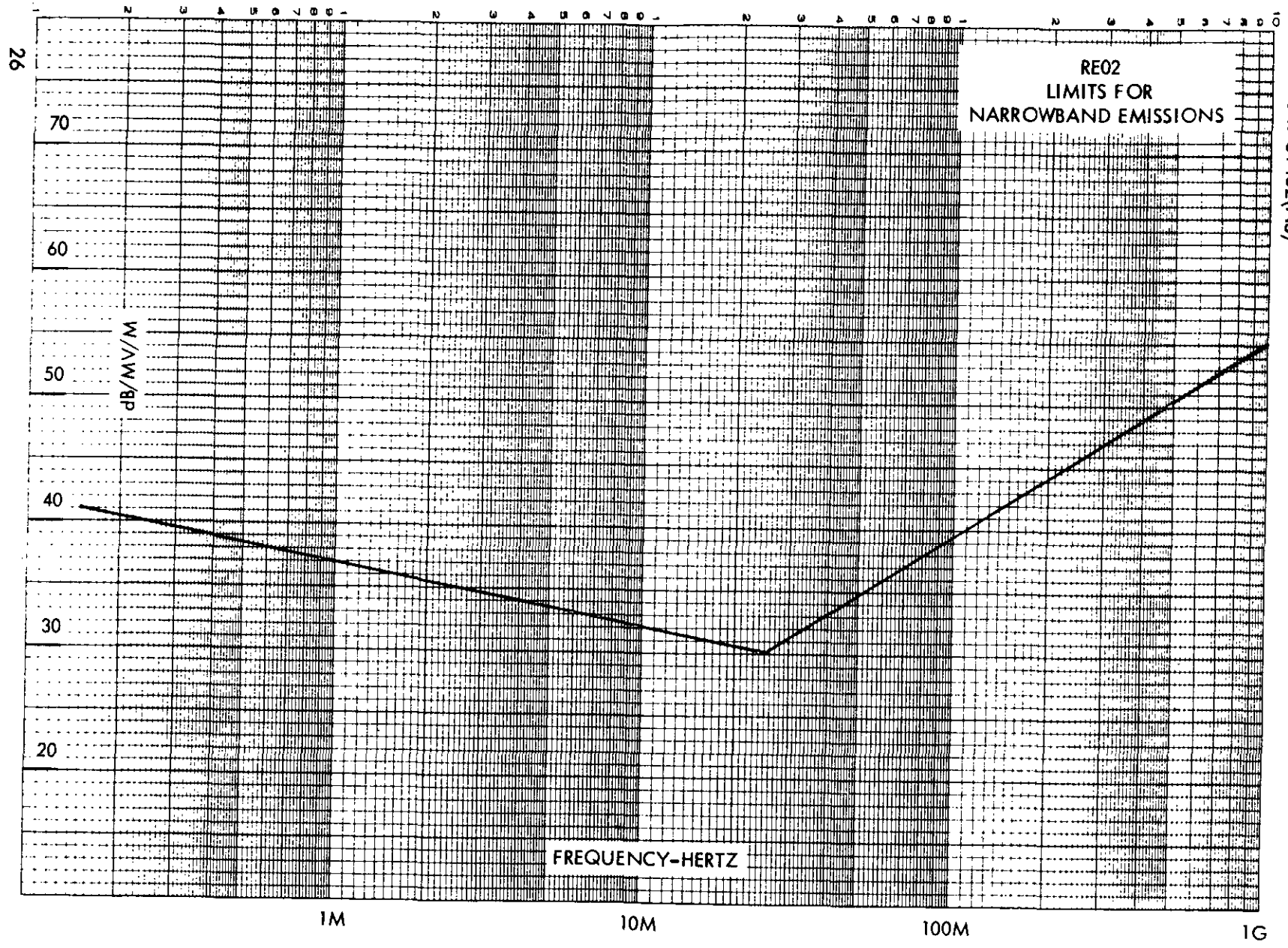
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(1) Manufactoring Run-In Tests

(2) Operational Tests

4.3.1 Manufacturing Run-In (MRI) Tests - Tests shall be conducted on each equipment. These tests shall consist of twenty (20) failure-free cycles (not consecutive) in accordance with (Test Level E) of MIL-STD-781. Prior to testing, each equipment shall undergo a visual inspection to determine that the material and workmanship requirements have been met. MRI tests are defined below.

Temperature:	0° -2° $+0^{\circ}$ to 40° $+5^{\circ}$ -0°C
Temperature Cycling:	Temperature cycling shall be time to stabilize at low temperature followed by time to stabilize at the high temperature, plus 2 hours.
Vibration:	2.2 G ± 10% peak acceleration at any nonresonant frequency between 20 and 30 Hz measured at a mounting point on the AN/SRC-40. The duration of vibration shall be at least 10 minutes per hour of AN/SRC-40 power-on time.
Equipment On/Off Cycling:	Equipment off during cooling cycle and on during heating cycle. A maximum of 4 hours per complete test cycle is to be performed.
Input Voltage:	115 VAC nominal, +5%, -2%
Input Voltage Cycling:	None

4.3.2 Operational Tests - Each equipment shall be functionally tested to determine compliance with operational requirements. This test shall be conducted after satisfactory completion of twenty (20) failure-free cycles, as defined in Paragraph 4.3.1. Sufficient characteristics shall be checked and adequate data shall be recorded to assure satisfactory equipment operation.

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4.4 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.5 Reconditioning of Tested Equipment - Any AN/SRC-40 which fails during preproduction testing shall be reconditioned by the supplier by replacing all damaged or defective items. After reworking, the supplier shall resubmit the equipment for acceptance. Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government inspector.

4.6 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

5.0 PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555. In the event the equipment is not covered in MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6.0 NOTES

6.1 Intended Use - The AN/SRC-40 is to be used as a digital data transmitter terminal for simultaneous UHF/FSK transmission of CAINS Alignment and Waypoint data to aircraft equipped with Link-4A airborne equipment equivalent to the AN/ASW-25B or AN/ASW-27B and for transmission of the ACLS data link signal when being operated as a single-channel transmitter.

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6.2 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.3 Definitions - Unless otherwise qualified in context, the following terms, as used herein, shall have the indicated meanings:

6.3.1 Approve - To have the authority to review and concur with or to suggest changes. Formal written approval is not intended unless specifically called out herein or in other contractual documents. Approval may be assumed by the requesting party, in the absense of a response to the contrary, 30 days after the date of the request. The affixing of an approval stamp or signature shall not reduce either party's responsibility for AN/SRC-40 performance as defined by this specification.

6.3.2 Contractor - The supplier of the AN/SRC-40.

6.3.3 ACLS - All-Weather Carrier Landing System.

6.3.4 CAINS - Carrier Aircraft Inertial Navigation System.

Project Number 7440-N095

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-R-81462(AS) RADIO SET AN/SRC-40 DIGITAL DATA TRANSMITTER TERMINAL

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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